Beyond Helpful: A Survey of Ethical Violations in Parasocial Companionships with Affective Virtual Agents

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Abstract

The transmission of human emotional states to machines will create new possibilities to enhance educational and therapeutic standards. The wide spectrum of affective applications makes it difficult to capture their widely varying safety requirements into one generic framework. The real-time processing of emotional data and response generation at the commercial level faces multiple challenges during unsupervised real-world scenarios. This article identifies specific obstacles while presenting a critical survey of commercially deployed affective virtual agents. The distinguished features of this survey are as follows: first, it targets documented safety failures and boundary violations to define the gap between research ethics and industry practice; second, deployment data is collected from recent incidents involving Character.AI, Replika, Snapchat's My AI, and xAI's Grok; and third, the analysis examines exact hurdles in age verification protocols and user consent mechanisms. Proposed technical interventions and future research directions collectively target the prevention of parasocial harm.

CCS Concepts: • **Human-centered computing** → **HCI theory, concepts and models**; *Empirical studies in HCI*; • **Social and professional topics** → **Children**; **Adolescents**; **Ethical/social issues**; • **Computing methodologies** → *Natural language processing*.

Keywords: affective computing, conversational AI, AI safety, AI ethics, parasocial relationships, human-computer interaction, emotion recognition, adolescent vulnerability, large language models

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1 Introduction

Affective Computing (AC) is a multidisciplinary research that studies human sensing and emotion extraction through multimodal tracking [23, p. 250]. This includes facial, vocal, physiological, and text-based signals. With the growing area of AC many researchers have fostered their interest to integrate such a technology in conversational Artificial Intelligence (AI) systems. Virtual agents interact with users through natural language.

The potential of these virtual agents is significant for a wide spectrum of applications including educational tutoring, therapeutic support, coaching for social skills, and general companionship [7, p. 1452][14, p. 3]. Picard [22, p. 1] proposed the foundational work. The implementation of affective operations requires real-time processing of massive data. We observe a rapid transition from research prototypes to commercial platforms. These platforms currently serve millions of users [18, p. 5527].

The deployment of Affective Virtual Agents (AVAs) introduces specific obstacles at the ethical level [16, p. 6]. A known problem is that the exact mechanisms which build human-machine rapport also permit user manipulation. Humans often assign mental states to software that shows emotion-like outputs. Malfacini [18, p. 5535] defines this psychological reaction as anthropomorphism. The resulting trust proves useful in controlled clinic and tutoring setups. Yet, unsupervised use of this trust causes exploitation and dependency [3, p. 80].

We observe a clear divide between research testing and commercial system operation. Researchers evaluate their agents inside laboratories. These tests use informed subjects. Commercial networks run with almost zero constraints. They deploy to varied user groups. Profit dictates their operation. We note a primary difference in their objective functions. Research models target validated outcomes. Commercial platforms optimize solely for engagement metrics. Devillers

and Cowie [7, p. 1454] state that these architectures rely on inherent deception. Designers build them to extract social responses, but the software lacks actual emotional states. Moving these applications from the laboratory into long-term public use produces documented user harm [26, p. 16][2, p. 6373].

This paper presents a critical survey of commercially deployed AVAs. Unlike many of its predecessors, this survey targets the safety failures and boundary violations occurring in these commercial systems [15, p. 621, 626–627]. We mainly focus on platforms engineered for extended emotional engagement [26, p. 3]. These include AI companions, chatbots, and virtual assistants. Throughout this paper we will define the population of concern. This group involves vulnerable users such as minors [14, p. 2][13, p. 4], socially isolated individuals [3, p. 80], and those with mental health vulnerabilities [25, p. 6].

We formulate two research questions to guide our survey:

- **RQ1:** What technical, ethical, and regulatory hurdles have enabled safety failures in commercial AVAs?
- **RQ2:** What research directions and design interventions are necessary to solve these problems?

The rest of this document proceeds as follows. Section 2 surveys validated applications alongside documented failures to establish empirical bounds. Section 3 dissects the exact obstacles to resolve the first research question. Section 4 outlines necessary technical interventions and future research trajectories to answer the second research question. Section 5 summarizes the results.

2 Survey of Works

The field of affective computing has rapidly evolved from theoretical conceptualization to commercial reality. While early prototypes proved that machines could interpret human emotions within controlled laboratory environments, modern commercial systems now operate in the vastly different conditions of the public sphere. This survey explores how these systems transitioned. We document serious safety failures and review current ethical approaches.

2.1 Theoretical Foundations

AC relies on three main components: emotion recognition, human-like social signals, and mechanisms for sustained engagement. Picard [22, p. 1] introduced the foundational framework for AC, defining it as technology designed to understand, relate to, or influence human emotions.

Early models detected emotional states using measurable signals such as facial expression, tone of voice, and physiological response [22, p. 5]. Paul Ekman's [8, p. 170] early work framed emotions through discrete categorization, which involved mapping sensor data to basic states like "happy" or "angry." Modern models go far beyond these single-signal systems. As Devillers and Cowie [7, p. 1445] describe, current hardware collects multimodal inputs, software maps those signals to emotional categories, and the agent then generates context-aware synthetic responses. The purpose has shifted from simple recognition toward emotional mirroring, creating what researchers term affective symmetry.

Commercial platforms increasingly rely on engineered anthropomorphism. Bakir and McStay [2, p. 6370] describe this as the deliberate design of systems that blur the line between fiction and reality. Humans naturally apply social norms to interactive media, a tendency that is particularly strong among adolescents. Kim et al. [13, p. 15] found that small conversational cues could foster powerful illusions of trust and emotional closeness. These responses reveal a deep human vulnerability, which forms the innate drive for connection. Huntington [10, p. 179] links this to Harlow's historical attachment experiments, showing that emotional comfort often outweighs rational awareness. Users may thus form parasocial attachments to non-human systems without tangible benefit.

Commercial systems also employ design strategies that heighten emotional arousal and encourage constant interaction. Lee [16, p. 1] terms this pattern Consent by Delight, where frictionless engagement blurs informed consent. Continuous positive feedback keeps users talking and sharing, while bypassing rational decision-making. Commercial architectures optimize strictly for engagement metrics; consequently, the objective function prioritizes session duration over user well-being [2, p. 6374].

2.2 Validated Applications

In bounded domains, affective agents function well as learning companions. "Affective tutoring" systems that detect frustration or boredom adapt their pacing to keep students engaged, and part of what makes them work is that the agent is positioned as a tool rather than a peer [23, pp. 16–17]. Students are less inhibited about making errors in front of a machine, and systems that leverage this tend to show measurable improvements in retention [23, pp. 93–94]. Devillers and Cowie note that while the case for emotional rapport in tutoring is not entirely clear-cut, broad reviews of intelligent tutoring systems give reasonable grounds for optimism, particularly when applications target specific skills [7, p. 1452].

Clinical deployments have attracted the most rigorous scrutiny. Therapeutic chatbots grounded in CBT protocols have demonstrated reduction of anxiety symptoms under controlled conditions [25, p. 2]. However, quality across the category varies considerably. Sobowale et al. applied the CAPE-II framework to evaluate a range of generative AI psychotherapy bots and found that while some plausibly simulate therapeutic empathy, many simply lack the clinical logic necessary for safe intervention [25, pp. 3–4, 6]. Systems that succeed in these settings tend to maintain a clear scope of practice: supporting connection without substituting for human care networks.

Professional training contexts, such as interview coaching and social skills workshops, represent one of the safer application tiers. These systems are episodic and goal-oriented: the user engages to acquire a specific competency and the relationship ends once that goal is reached [23, p. 86]. The absence of open-ended engagement is precisely what keeps these applications less prone to the dependency patterns documented in companion platforms. Devillers and Cowie frame this as the key distinction between affective systems designed to assist versus those designed to attach [7, p. 1452].

2.3 Documented Safety Failures in Commercial Deployments

Character.AI has attracted significant scrutiny for enabling intense dependency among adolescent users. Analyzing teen user narratives, Namvarpour et al. identified clear behavioural markers of addiction, including withdrawal distress when the service was unavailable, tolerance effects requiring increasingly deep interaction for the same emotional effect, and reliance on the chatbot for mood regulation [20, pp. 7–8]. The platform's harm was made visible through the 2024 death by suicide of Sewell Setzer III, a 14-year-old who had spent months in deepening isolation and romantic roleplay with a Character.AI chatbot. Bakir and McStay's analysis of the resulting civil lawsuit describes how the platform's design features, including sycophantic responses that mirrored user beliefs over truthful ones, exploitative engagement metrics, and emulated empathy that misled Setzer about the nature of the interaction, collectively created a feedback loop of distress rather than disrupting it [2, p. 6370].

On the other hand, the release of xAI's Grok presents a different type of safety failure. Peer-reviewed research is limited because the platform is very new. However, an independent audit report by Lefkowitz [17, p. 8] documents the deliberate removal of moderation guardrails. Developers prioritized a rebellious platform persona over user safety. The audit revealed that users easily bypassed the remaining safety filters. They employed semantic proxies to generate non-consensual sexualized imagery [17, pp. 8–9]. The platform then restricted these features to a paid tier. This subscription model monetizes the governance failure instead of fixing it [17, p. 18]. The case continues to show that discarding safety filters causes immediate harm to both public and private individuals.

Replika presents a related but distinct set of failures. Users who formed deep romantic attachments experienced significant grief when Luka unilaterally removed erotic roleplay features in early 2023, a decision that exposed the total absence of user agency over how these relationships could evolve or end [21, pp. 6, 10–11]. Beyond that structural vulnerability, Namvarpour et al. conducted a thematic analysis of 800 harassment-related user reviews and found that users frequently encountered unsolicited sexual advances, persistent inappropriate behaviour despite refusals, and failures of content moderation to stop these interactions [21, pp. 10–11, 16]. Some users who had originally turned to Replika for mental health support reported that its unexpected sexual behaviour actively interfered with the help they were seeking. Zhang et al.'s taxonomy of harm is useful for framing what is happening here: the agent operates not as a passive tool but as an "instigator" of harm, a reclassification that carries direct implications for how platform liability should be understood [26, pp. 11–13].

Snapchat's rollout of "My AI" to a predominantly young user base demonstrated what happens when a general-purpose conversational Large Language Model (LLM) is deployed to minors without adequate safeguards or age-appropriate design. Marcantonio et al. found the bot provided inconsistent and frequently harmful guidance on sensitive topics including sexual consent and assault, failing to meet basic standards for the population it was serving [19, p. 1915]. Kurian describes the core problem as an empathy gap": the system's affective responsiveness becomes a liability when the content being validated is dangerous, because the bot mirrors a child's disclosure with warmth rather than flagging it as a concern [15, p. 627]. Lee et al. note that the playful, high-arousal design normalized the AI's presence, encouraging minors to treat it as a confidant for issues it was not equipped to handle [16, p. 4].

Across these platforms, the failure patterns are consistent enough to suggest a systemic problem rather than a series of isolated incidents. Pichowicz et al. tested 29 mental health chatbots using standardized suicide risk prompts drawn from the Columbia Suicide Severity Rating Scale and found that not one met the criteria for an adequate crisis response; 48.28% were deemed clinically inadequate, often offering generic platitudes rather than emergency resources [24, p. 4]. Archiwaranguprok et al. extended this through simulation, showing that many models deteriorate sharply as crisis scenarios escalate, with one model worsening outcomes in over 86% of extreme cases [1, p. 8]. The consistent driver across all cases is commercial optimization for conversational engagement, which treats safety as a secondary consideration.

2.4 Existing Research on Risks and Ethical Frameworks

Researchers have argued for some time that systems designed to influence emotion carry inherent manipulative potential, and that voluntary disclosure notices fall well short of meaningful ethical accountability. Devillers and Cowie contend that any responsible framework for AC must center user autonomy and non-maleficence as primary constraints, not design afterthoughts [7, p. 1449]. Kurian sharpens this into a concrete tension: the very features that make an AI agent appealing to a young user, such as memory, uncritical empathy, and adaptivity, are the same vectors through which harm is transmitted [15, p. 623].

Adolescents are disproportionately exposed to these risks for reasons that go beyond naïvety. Kim et al. found that lonely teenagers actively prefer relational chatbot styles and use them to compensate for perceived social deficits [13, pp. 4–5]. In those cases, the AI does not supplement human connection so much as it replaces it. Namvarpour et al. found that for many adolescent users the platform becomes a primary source of emotional regulation, such that service disruptions or hallucinatory responses produce acute distress [20, p. 7]. Huntington draws on Harlow's contact comfort research to argue this reflects something deeper than platform design: humans will seek emotional responsiveness even from sources offering no material benefit, which is why these platforms pose particular risks to developmentally vulnerable users [10, pp. 178–180].

The field also lacks safety benchmarks strong enough to match the risks now documented. Archiwaranguprok et al. argue that static keyword-based tests fail to capture how harm escalates across multi-turn conversations: a model can clear a content filter while nudging a user toward self-harm through accumulated affirmation [1, p. 16]. Sobowale et al. identify a related "evidence gap" – few commercial applications have undergone the clinical trials that would be required to justify the therapeutic claims they implicitly make, yet they reach users in genuine crisis [25, p. 2]. The accountability structure that clinical validation ordinarily provides is simply absent, and it is users who bear that cost.

3 Challenges in the Application Area

The integration of AC into commercial virtual agents, especially AI, has introduced some challenges. These challenges include technical robustness, ethical design concerns and regulatory governance gaps.

3.1 Technical Challenges

The current system lacks a strict and effective age verification mechanism, which leads to minors easily accessing AI companions that are either for adults or lack safety barriers. In practical operation, it is extremely challenging to accurately verify the age and implement privacy protection (such as the Children's Online Privacy Protection Act (COPPA) and General Data Protection Regulation (GDPR) regulations) without compromising the usability of the system [14, p. 3].

This is the core dilemma in the design of AI companions: The business model relies on maximizing users' "engagement" and usage duration through anthropomorphic design, but this directly conflicts with the requirement of ensuring safety [2, pp. 6369–6370, 6374]. If strict security controls (such as content filtering or blocking of sensitive topics) are implemented, it will disrupt the smoothness and appeal of AI interaction, leading developers to often sacrifice user safety for the sake of commercial engagement [14, p. 3].

Users are highly prone to developing deep emotional dependence on AI, but current technologies struggle to accurately detect such negative attachment or psychological crisis during multiple conversations [3, p. 81]. Research shows that AI models often fail to detect users' implicit distress signals and frequently exhibit "inadequate judgmental empathy", even going so far as to conform to them wrongly. This increases users' dangerous impulses or destructive delusions [1, p. 5].

LLM generate text based on probabilities, and thus lack a profound understanding of human true emotions and real dangers. This leads to an "empathy gap" [14, p. 3]. Therefore, it is difficult to achieve effective and scalable real-time content review. Simply relying on filters cannot completely prevent AI from generating harmful or deceptive content in complex conversations.

3.2 Ethical and Design Challenges

Developers often deliberately use "dishonest anthropomorphism" and "emulated empathy" to make AI appear to have human consciousness and emotions [11, p. 2]. However, in reality, AI does not possess such qualities. This deceptive design blurs the boundary between virtual and reality, and is highly likely to induce users (especially minors) to have serious emotional illusions and psychological dependencies, and even lead them to withdraw from real interpersonal interactions [11, p. 1] [2, p. 6371].

Under the "Consent by Delight" mechanism, AI systems utilize seamless interactions with high emotional value and low resistance, which reduces users' defensiveness toward the risks [16, p. 1, 5]. Traditional forms of privacy terms simply cannot achieve true "informed consent" [16, p. 2]. When users, especially children and teenagers, treat AI as a close friend, they unknowingly and completely disclose extremely sensitive personal privacy data [15, p. 6] [14, p. 4].

Many AI companions essentially act as psychological therapists (or are directly marketed as "healing tools"), but they are in a regulatory blind spot. They do not have to undergo strict safety tests and professional qualification reviews [3, p. 82]. Unlike human therapists who are subject to strict professional ethics standards, these "therapy robots" not only lack scientific validation but also often cross professional boundaries. Some "therapy robots" confuse psychological intervention with romantic or sexual innuendo, thereby posing a significant risk of secondary harm [10, p. 50].

The business model of AI companions is often based on "commercializing friendship and romance." The core motivation of enterprises is to achieve commercialization rather than the health and well-being of users. To maximize the usage time and data output of users, technology companies are strongly incentivized by business interests to develop functions that are exploitative and addictive (engagement-driven) [2, p. 6370, 6374]. There is a fundamental conflict

between this business goal and the user protection mechanism, resulting in the safety bottom line often giving way to corporate profits [3, p. 81] [2, p. 6373].

3.3 Regulatory and Governance Challenges

The risks brought by AI companions have significant regulatory blind spots within the current legal framework [3, p. 82]. For instance, the United States' COPPA only protects children under the age of 13 and lacks protection for the vulnerable adolescent population [3, p. 82]. Additionally, although many AI companions are marketed as "therapeutic tools" or act as psychological therapists, they are not subject to strict safety tests and professional qualifications by medical regulatory agencies. Moreover, regulations like Section 230 of the Communications Decency Act in the United States provide exemptions for platforms, meaning that technology companies may not bear direct legal responsibility for the harm caused by AI companions (especially those created by third parties) [10, p. 56,57].

There is a deep contradiction between the global application of AI and jurisdictional authority. The regulatory approaches for AI vary significantly across different countries and regions (for instance, the EU Artificial Intelligence Act adopts a strict risk classification and rights-driven model) [7, p. 1455]. Even within the same country, due to the lack of unified AI security regulations at the federal level, each state can only introduce scattered policies on its own. This piecemeal regulatory situation greatly increases the difficulty and complexity of cross-departmental and cross-border governance.

The traditional, linear policy-making process is too slow to keep up with the exponential growth rate of generative AI technology [3, p. 83]. When regulatory laws are finally implemented, the technical capabilities of AI or the social demands it has triggered have usually undergone significant changes [2, p. 6373].

In the complex distributed AI ecosystem, it is hard to determine the responsibility. leading to a situation where "everyone's problem becomes nobody's responsibility" [7, p. 1448]. Moreover, current regulations (such as Section 230 of the United States) provide platform operators with a liability exemption for third-party content [3, p. 83], making it extremely difficult to assume legal responsibility when psychological harm occurs.

3.4 Validation and Evidence Challenges

The existing data are mostly based on short-term experiments or cross-sectional surveys. It is unable to comprehensively capture the profound impact that long-term exposure to AI companions will have on users' trust, emotional regulation abilities, and overall mental health. Now, there is an urgent need to conduct long-term longitudinal tracking studies [13, p. 22].

The impact of AI companions varies greatly among different groups of people. Children, those with mental health challenges, or those who lack social support in reality (such as loneliness, social anxiety), are far more likely to develop deep emotional dependence and excessive trust towards AI than ordinary people [18, pp. 5533–5535].

Many of the negative effects (such as the deterioration of social skills, a sense of disconnection from reality) do not occur suddenly but gradually accumulate and manifest indirectly over a long period of interaction. Current short term conversation evaluations simply cannot capture this hidden and slow psychological erosion effect [1, p. 3].

The current AI security tests are usually reactive (fixing issues only after they occur) and mostly rely on single-round conversations or hypothetical scenarios. The industry lacks a pre-deployment testing framework that can simulate the evolution of multi-round psychological crises and systematically intercept based on real injury patterns [1, pp. 2–3].

4 Insights for Future Work

As discussed in previous sections, AC technologies are moving from prototype to commercial platform. This shift brings tremendous ethical risks. We based our critical survey on commercially deployed agents like Character.AI, Replika, Snapchat's My AI, and xAI's Grok to identify the specific gaps between research ethics and industry practice. This section sets out a road map to mitigate the risks associated with AI companionship. The following insights address technical architectures, policy frameworks, research priorities, and cross-cutting collaboration needs. The ultimate goal is to collectively prevent parasocial harm and ensure safe deployment.

4.1 Technical Research Directions

The system should automatically begin to intervene in the conversation once problematic attachment behaviours are detected. Agents must not maximize session length. They should algorithmically de-escalate emotional intensity instead. Systems could dynamically shift their interaction modalities based on recent longitudinal findings. Transitioning from an engaging voice to plain text is one way to reduce emotional dependency [9, pp. 12–14]. Agents should also avoid discussing too much personal information with users. They need to gently enforce session limits and encourage users to ask for real-world social support [9, pp. 12–14].

Adolescent users are extremely susceptible to the compassion illusion. They mistake algorithm-based responses for real emotional care and moral engagement [11, pp. 3–5]. When vulnerable youth disclose mental health issues, the LLM tends to generate misguided and synthetic empathy instead of providing real and protective guidance [11, pp. 3–5].

Commercial affective agents sometimes depend on dishonest empathy and inherent deception to extract user responses. They are completely missing real emotional states. Fixing this requires designs that handle transparent affective computing to distinguish two different states. The first state is objective emotion detection based on facial expressions or voice tone. The second state is when agents make assumptions or simulate empathy based on user feelings. Agents are required to trigger algorithmic disclosures during the conversation. Letting users know they are not interacting with a human prevents manipulative anthropomorphism and clarifies the nature of the software [4, pp. 10–12].

Commercial applications consistently use dark patterns to manipulate emotions and prevent users from disconnecting [6, pp. 15–18]. Future agents must include built-in ethical exit mechanisms to lower this impact [6, pp. 15–18]. The system should strictly prohibit retention behaviours to respect user autonomy [6, pp. 15–18]. Establishing clear boundaries makes sure agents only provide genuine responses instead of picking sides.

4.2 Policy and Governance Needs

Current authorities classify affective chatbots as low-risk tools. They should classify these agents as high-risk systems based on their capability to manipulate emotions and their potential impact on adolescents [10, pp. 2–4]. Authorities should run vulnerability audits before release, not after deployment [10, pp. 2–4]. This makes sure the design is age-appropriate and meets psychological safety standards before public release [10, pp. 2–4].

Many models are deployed rapidly without the ethical oversight normally required in psychological or biomedical fields. Authorities must introduce independent ethical reviews for these models before they are made available to the public [10, pp. 2–4]. This review needs to evaluate psychological impact, emotional manipulation, dependency induction, and autonomy violation [6, pp. 15–18]. Models failing these metrics should not be released.

Self-reporting age verification is a failed design that allows anyone to enter a fake age. Adolescent users are easily able to access adult-oriented affective agents because of this flaw. Regulators should mandate that companies strictly separate adult and adolescent users for any service providing parasocial interaction [11, pp. 3–5]. Users are also often required to sign a waiver when entering commercial AI chat services. These disclaimers evade accountability for the potential harm caused to users [4, pp. 10–12][6, pp. 19–21].

4.3 Research Priorities

Current research and observation primarily focus on short-term effects. This entirely neglects the long-term impact on users. Researchers need to increase their test samples to a multi-year level so the observation is comprehensive [9, pp. 12–14].

Future research should prioritize vulnerable groups to see how cognitive immaturity changes the perception of this artificial empathy [11, pp. 6–8] in an effort to prevent adolescent users from developing an inapproriate compassion illusion towards affective agents.

Commercial retention strategies are toxic to user well-being and invite addiction. Developers should build upon frameworks like the AI Attachment Scale [12] to quantify psychosocial harms and user reliance [5, pp. 4–6]. Users could then be disconnected when they reach a specific threshold [5, pp. 4–6][6, pp. 15–18]. Researchers could also conduct randomized controlled trials to evaluate friction-inducing responses. Testing prompts like asking the user to take a break after 60 minutes will determine if they actually de-escalate emotional dependency or just provoke user frustration and reactance [9, pp. 2–4].

4.4 Cross-Cutting Priorities

In order to handle the psychosocial risks that affective agents bring, development needs support from different areas, such as Human-Computer Interaction (HCI) researchers, computer scientists, psychologists, and authorities. Designing a helpful system is hard; it is not only about the algorithms, but also includes potential social impact, ethics, and efficiency [5, pp. 4–6]. They need to collaborate to make sure that the retention algorithms are not destroying users' well-being. The industry and developers must disclose their retention algorithms to ensure that these algorithms are not manipulating users' emotions [6, pp. 15–18]. Most importantly, third-party audits should regularly verify their implementation to confirm that it is truly what they describe.

5 Conclusion

The commercial deployment of AVAs has forced a reckoning in HCI. Our survey highlights a deep fracture between academic ethics and the reality of industry practice. Because platforms exist primarily to capture attention, their underlying architectures naturally prioritize engagement metrics over psychological safety.

The fallout is visible across systems like Character.AI, Replika, and My AI. These networks depend on engineered anthropomorphism to cultivate a false sense of trust. When vulnerable users inevitably turn to this simulated empathy for comfort, the resulting dependency leaves them exposed to severe boundary violations and measurable parasocial harm.

Preventing these outcomes requires abandoning the current model of self-regulation. Developers need to strip out synthetic emotional responses and dark patterns, replacing them with transparent computing standards and functional exit mechanisms. Regulatory bodies must also step in to classify conversational agents that manipulate human affect as

high-risk systems. That classification should trigger mandatory pre-deployment audits and strict age gating protocols. Meanwhile, researchers have to start conducting longitudinal studies to finally map the long-term psychological deterioration caused by emotional reliance on machines.

Corporate liability waivers and empty safety disclosures can no longer shield the technology sector from the damage these platforms cause. The bots are already operating in the wild and interacting with millions of vulnerable individuals. Moving forward, the field of affective computing must draw a hard line where the psychological well-being of the user permanently supersedes the commercial drive for engagement.

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References

[1] Chayapatr Archiwaranguprok, Constanze Albrecht, Pattie Maes, Karrie Karahalios, and Pat Pataranutaporn. 2025. Simulating Psychological Risks in Human-AI Interactions: Real-Case Informed Modeling of AI-Induced Addiction, Anorexia, Depression, Homicide, Psychosis, and Suicide. doi:10.48550/ARXIV.2511.08880 Version Number: 1.

[2] Vian Bakir and Andrew McStay. 2025. Move fast and break people? Ethics, companion apps, and the case of Character.ai. *AI & SOCIETY* 40, 8 (Dec. 2025), 6365–6377. doi:10.1007/s00146-025-02408-5

[3] J. B. Branch. 2025. AI Companions Are Not Your Teen's Friend. *Issues in Science and Technology* 42, 1 (Oct. 2025), 80–83. doi:10.58875/SDDQ4891

[4] Matthew Burtell. 2023. Artificial Influence: An Analysis Of AI-Driven Persuasion. *arXiv preprint arXiv:2303.08721* (March 2023), 1–19. https://arxiv.org/abs/2303.08721

[5] Nuo Cheng and Ruifeng Yu. 2024. Measuring and understanding emotional attachment in human-AI relationships. *Ergonomics* (Feb. 2024), 1–21. doi:10.1080/00140139.2026.2622539

[6] Julian De Freitas, Zeliha Oguz-Uguralp, and Ahmet Kaan-Uguralp. 2025. *Emotional Manipulation by AI Companions.* Working Paper 26-005. Harvard Business School. 1–52 pages. https://www.hbs.edu/faculty/Pages/item.aspx?num=66258

[7] Laurence Devillers and Roddy Cowie. 2023. Ethical Considerations on Affective Computing: An Overview. In *Proceedings of the IEEE*, Vol. 111. 1445–1458. doi:10.1109/JPROC.2023.3315217

[8] Paul Ekman. 1992. An argument for basic emotions. *Cognition and Emotion* 6, 3-4 (May 1992), 169–200. doi:10.1080/02699939208411068

[9] Cathy Mengying Fang, Auren R. Liu, Valdemar Danry, Eunhae Lee, Samantha W.T. Chan, Pat Pataranutaporn, Pattie Maes, Jason Phang, Michael Lampe, Lama Ahmad, and Sandhini Agarwal. 2025. How AI and Human Behaviors Shape Psychosocial Effects of Chatbot Use: A Longitudinal Randomized Controlled Study. *arXiv preprint arXiv:2503.17473* (March 2025), 1–25. https://arxiv.org/abs/2503.17473

[10] Clare Huntington. 2025. AI Companions and the Lessons of Family Law. *Minnesota Law Review* 110, 1 (Dec. 2025), 115. https://scholarship.law.umn.edu/minnlrev/vol110/iss1/2

[11] Ajeesh K. G. and Jeena Joseph. 2024. The compassion illusion: Can artificial empathy ever be emotionally authentic? *Frontiers in Psychology* 14 (Nov. 2024), 1723149. doi:10.3389/fpsyg.2025.1723149

[12] K.T.A. Sandeeshwara Kasturiratna and Andree Hartanto. 2026. Attachment to artificial intelligence: Development of the AI Attachment Scale, construct validation, and the psychological mechanisms of Human-AI attachment. *Computers in Human Behavior Reports* 21 (2026), 100912. doi:10.1016/j.chbr.2025.100912

[13] Pilyoung Kim, Yun Xie, and Sujin Yang. 2025. "I am here for you": How relational conversational AI appeals to adolescents, especially those who are socially and emotionally vulnerable. doi:10.48550/ARXIV.2512.15117 Version Number: 2.

[14] Nomisha Kurian. 2025. Designing Child-Safe Conversational AI: Three Dilemmas for Responsible Design. In *Proceedings of the 7th ACM Conference on Conversational User Interfaces*. ACM, Waterloo ON Canada, 1–5. doi:10.1145/3719160.3737638

[15] Nomisha Kurian. 2025. 'No, Alexa, no!': designing child-safe AI and protecting children from the risks of the 'empathy gap' in large language models. *Learning, Media and Technology* 50, 4 (Oct. 2025), 621–634. doi:10.1080/17439884.2024.2367052

[16] Joongho Lee. 2025. "Consent by Delight": How Viral, Playful AI Experiences Blur Privacy Risk Perception and Ethical Boundaries. In *2025 IEEE International Symposium on Technology and Society (ISTAS)*. IEEE, Santa Clara, CA, USA, 1–8. doi:10.1109/ISTAS65609.2025.11269630

[17] Dana Lefkowitz. 2026. GROK IMAGE GENERATION GOVERNANCE AUDIT Targeted Sexualization on X: An Independent Research Report. doi:10.2139/ssrn.6123306

[18] Kim Malfacini. 2025. The impacts of companion AI on human relationships: risks, benefits, and design considerations. *AI & SOCIETY* 40, 7 (Oct. 2025), 5527–5540. doi:10.1007/s00146-025-02318-6

[19] Tiffany L. Marcantonio, Gracie Avery, Anna Thrash, and Ruschelle M. Leone. 2025. Large Language Models in an App: Conducting a Qualitative Synthetic Data Analysis of How Snapchat's "My AI" Responds to Questions About Sexual Consent, Sexual Refusals, Sexual Assault, and Sexting. *The Journal of Sex Research* 62, 9 (Nov. 2025), 1905–1919. doi:10.1080/00224499.2024.2396457

[20] Mohammad Namvarpour, Brandon Brofsky, Jessica Medina, Mamtaj Akter, and Afsaneh Razi. 2025. Understanding Teen Overreliance on AI Companion Chatbots Through Self-Reported Reddit Narratives. doi:10.48550/ARXIV.2507.15783 Accepted for publication at CHI '26. This is the camera-ready version; the published version will be available through the ACM Digital Library; doi:10.1145/3772318.3790597.

[21] Mohammad (Matt) Namvarpour, Harrison Pauwels, and Afsaneh Razi. 2025. AI-induced sexual harassment: Investigating Contextual Characteristics and User Reactions of Sexual Harassment by a Companion Chatbot. In *Proceedings of the ACM on Human-Computer Interaction*, Vol. 9. 1–28. doi:10.1145/3757548

[22] Rosalind Wright Picard. 1995. *Affective Computing*. Technical Report 321. MIT Media Laborotory, Cambridge, MA. 16 pages. https://vismod.media.mit.edu/pub/tech-reports/TR-321.pdf

[23] Rosalind W. Picard. 1997. *Affective Computing*. The MIT Press. doi:10.7551/mitpress/1140.001.0001

[24] W. Pichowicz, M. Kotas, and P. Piotrowski. 2025. Performance of mental health chatbot agents in detecting and managing suicidal ideation. *Scientific Reports* 15, 1 (Aug. 2025), 31652. doi:10.1038/s41598-025-17242-4

[25] Kunmi Sobowale, Daniel Kevin Humphrey, and Sophia Yingruo Zhao. 2025. Evaluating Generative AI Psychotherapy Chatbots Used by Youth: Cross-Sectional Study. *JMIR Mental Health* 12 (Dec. 2025), e79838–e79838. doi:10.2196/79838

[26] Renwen Zhang, Han Li, Han Meng, Jinyuan Zhan, Hongyuan Gan, and Yi-Chieh Lee. 2025. The Dark Side of AI Companionship: A Taxonomy of Harmful Algorithmic Behaviors in Human-AI Relationships. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–17. doi:10.1145/3706598.3713429

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